

NLP - Tutorial 1

1 Setting-up Python, Keras, and TensorFlow with Anaconda Distribution

1.1 Setting up Anaconda on Your Computer

<https://anaconda.com>

Anaconda is a powerful distribution for Python and data science packages. To get started, follow these steps :

1. Accept the license and agreement on the Anaconda website.
2. Update conda to ensure you have the latest version :

```
conda update -n base -c defaults conda
```

1.2 Testing the Installation

To verify that Anaconda is installed correctly, open a terminal and run the following command :

```
conda
```

This will display information about conda, confirming that it is installed.

1.3 Setting-up TensorFlow and Keras

Follow these steps to install TensorFlow and Keras within an Anaconda environment :

1. Create a new environment (e.g., 'name-environment') using the following command :

```
conda create -n name-environment
```

Confirm by answering 'yes' when prompted.

2. Activate your environment :

```
conda activate name-environment
```

3. Install TensorFlow and Keras using the following command :

```
conda install keras tensorflow
```

1.4 Checking the Installation

You can verify the installation by printing the versions of TensorFlow and Keras using Python. Run the following commands :

```
python -c 'import tensorflow as tf; print(tf.__version__)'  
python -c 'import keras; print(keras.__version__)'
```

This will display the versions of TensorFlow and Keras in your Anaconda environment.

2 Setting-up spaCy

2.1 Installation of spaCy with pip

You can install spaCy with the following command :

```
pip install spacy
```

2.2 Updating spaCy version to the latest

To update spaCy to the latest version, use the following command :

```
pip install -U spacy
```

2.3 Installation of spaCy with conda

Conda support for spaCy is provided by the conda community. To install spaCy with conda, use the following command :

```
conda install -c conda-forge spacy
```

2.4 Checking the version you have

To check the version of spaCy you have installed, run the following command :

```
python -m spacy info
```

2.5 Installation of spaCy on Windows

If you are using a Windows system, you will need to install a version of Visual C++ Build Tools or Visual Studio Express that matches your Python distribution. After that, you should be able to import spaCy into your Python shell with the command :

```
import spacy
```

3 Installing Language Models

You can install spaCy models just like any other Python module and include them as part of your Python application. They are properly versioned and can be added to your ‘requirements.txt’ file as dependencies. You can install models from a download URL or a local directory manually or via pip. The model data can be placed anywhere on your local filesystem.

To download a model, you can use spaCy’s download command. The ‘download’ command finds the most compatible model for your spaCy version, downloads it, and installs it. This is the easiest way to install a model :

```
$ python -m spacy download en_core_web_md
```

The above command selects and downloads the most compatible version of the specified model for your local spaCy version.

Another option is to download models separately for different languages :

```
$ python -m spacy download en
$ python -m spacy download fr
```

Once you have downloaded the models, you can load them using spaCy’s ‘load()’ method. Here’s what we have done so far :

```
$ pip install spacy
$ python -m spacy download en_core_web_md
import spacy
nlp = spacy.load('en_core_web_md')
doc = nlp('your doc text')
```

4 Visualization with displaCy

Visualization is a crucial tool for data scientists, as it allows for effective communication of concepts to both technical and non-technical audiences. Visualizing language data is particularly useful for identifying patterns in data quickly.

There are several Python libraries and plugins for visualization, such as Matplotlib, Seaborn, TensorBoard, and more. spaCy, being an industrial-strength library, comes with its own visualizer called ‘displaCy’. In this subsection, you’ll learn how to set up a ‘displaCy’ server on your machine, use it in a Jupyter notebook, and integrate it into a web application. You’ll also discover how to export the graphics you create as image files, customize visualizations, and make manual annotations without creating a ‘Doc’ object. We’ll start by exploring the easiest way, which is using ‘displaCy’'s interactive demo.

5 Getting Started with displaCy

To get started with ‘displaCy’, you can use the interactive demo available at <https://explosion.ai/demos/displacy>.

Follow these steps :

1. Navigate to the provided URL.
2. Enter your text in the "Text to parse" box.
3. Click the search icon on the right to generate the visualization.

The visualizer performs two syntactic parses, POS tagging, and a dependency parse on the submitted text to visualize the sentence’s syntactic structure. For now, you don’t need to worry about how POS tagging and dependency parsing work, as we will explore them in upcoming chapters. Just think of the result as a representation of the sentence’s structure.

In the demo, you’ll notice two checkboxes : "Merge Punctuation" and "Merge Phrases." "Merge Punctuation" combines punctuation tokens with the previous token to create a more compact visualization, which is especially useful for long documents.

The "Merge Phrases" option combines adjectives and nouns into one phrase, resulting in a more concise dependency tree. If you don’t merge, adjectives and nouns will be displayed individually. This feature is handy for visualizing long sentences with many noun phrases.

5.1 Entity Visualizer

displaCy's entity visualizer highlights the named entities in your text. The online demo is available at <https://explosion.ai/demos/displacy-ent/>.

Named entities can be thought of as proper nouns for important entities such as people's names, company names, dates, city and country names, and more.

The online demo functions similarly to the syntactic parser demo. Enter your text into the textbox and click the search button. Here is an example :

5.2 Visualizing within Python

With the introduction of the latest version of spaCy, the 'displaCy' visualizers are integrated into the core library. This means that you can start using 'displaCy' immediately after installing spaCy on your machine. Let's go through some examples.

The following code segment is the easiest way to start 'displaCy' on your local machine :

```
import spacy
from spacy import displacy
nlp = spacy.load('en\_core\_web\_md')
doc = nlp('your text.')
displacy.serve(doc, style='dep')
```

The following is what we did :

1. We import spaCy.
2. Following that, we import 'displaCy' from the core library.
3. We load the English model that we downloaded in the "Installing spaCy's statistical models" section.
4. Once it is loaded, we create a 'Doc' object to pass to 'displaCy'.
5. We then start the 'displaCy' web server by calling 'serve()'.
6. We also specify 'dep' as the style parameter to visualize the dependency parsing result.

After running this code, you should receive a response from 'displaCy'. Check it.

If you wish to use another port or you get an error because the port 5000 is already in use, you can use the port parameter of displaCy with another port number.

Replacing the last line of the preceding code block with the following line will suffice :

```
displacy.serve(doc, style='dep', port= '5001')
```

Here, we provide the port number 5001 explicitly. In this case, displaCy will render the graphics on `http://0.0.0.0:5001`.

Generating an entity recognizer is done similarly. We pass `ent` to the style parameter instead of `dep` :

```
import spacy
from spacy import displacy
nlp = spacy.load('en_core_web_md')
doc = nlp('Bill Gates is the CEO of Microsoft.')
displacy.serve(doc, style='ent')
```

5.3 Using displaCy in Jupyter Notebooks

Jupyter notebook is an essential part of daily data science work. Fortunately, 'displaCy' can detect whether you are currently coding in a Jupyter notebook environment and return markup that can be directly displayed in a cell.

If you don't have Jupyter notebook installed on your system but wish to use it, you can follow the instructions at

<https://test-jupyter.readthedocs.io/en/latest/install.html>.

In a Jupyter notebook, you can use the `'render()'` function instead of `'serve()'`. The rest of the code remains the same. You can type or paste the following code into your Jupyter notebook :

```
import spacy
from spacy import displacy
nlp = spacy.load('en_core_web_md')
doc = nlp('Bill Gates is the CEO of Microsoft.')
displacy.render(doc, style='dep')
```

Practice by considering some sentences.